

FOSSEE — eSim
Semester Long Internship Screening

SEMESTER LONG INTERNSHIP – AUTUMN 2026
INTERNSHIP SCREENING TASK REPORT
Task 7 – PCB Design using eSim

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PCB RF LOW NOISE AMPLIFIER (LNA)

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1.Summary

This report documents the design of a 2.4 GHz RF low-noise amplifier (LNA) front-end board with an integrated, regulated power and bias stage, produced as a screening submission for the FOSSEE Autumn Internship 2026 program run by IIT Bombay (Task 7 – PCB Design using eSim).

The goal of the board is to take a proven, datasheet-based RF amplifier IC and build the supporting hardware around it end to end with RF input/output matching, DC blocking, a clean regulated bias supply, and a status indicator. By schematic representation, mapping to real footprints and laying out a manufacturable 4-layer PCB.

2. Understanding the Circuit

Power Regulation Section

The board is powered from a 9–12V DC barrel jack. A 1N5819 Schottky diode (D1) protects the circuit against reverse-polarity connection. An AMS1117-5.0 low-dropout voltage regulator (U2) then produces a clean, regulated 5V rail, with bulk decoupling capacitors (C5, C6) placed at its input and output pins to suppress supply noise before it reaches the RF stage.

RF Signal Chain and Bias Network

The RF signal path runs from SMA connector J1 through DC-blocking capacitor C1, into the SPF5189Z LNA (U1), through a second blocking capacitor C2, and out through SMA connector J2. Because U1 combines its RF output and DC bias supply on the same pin, an RF choke inductor (L1, 47 nH) forms a bias tee, that is, it presents high impedance to the RF signal (keeping it out of the DC supply) while passing DC bias current freely to the amplifier. Local decoupling capacitors C3 and C4 sit directly at this bias node to filter high-frequency ripple to the IC

External Connectivity

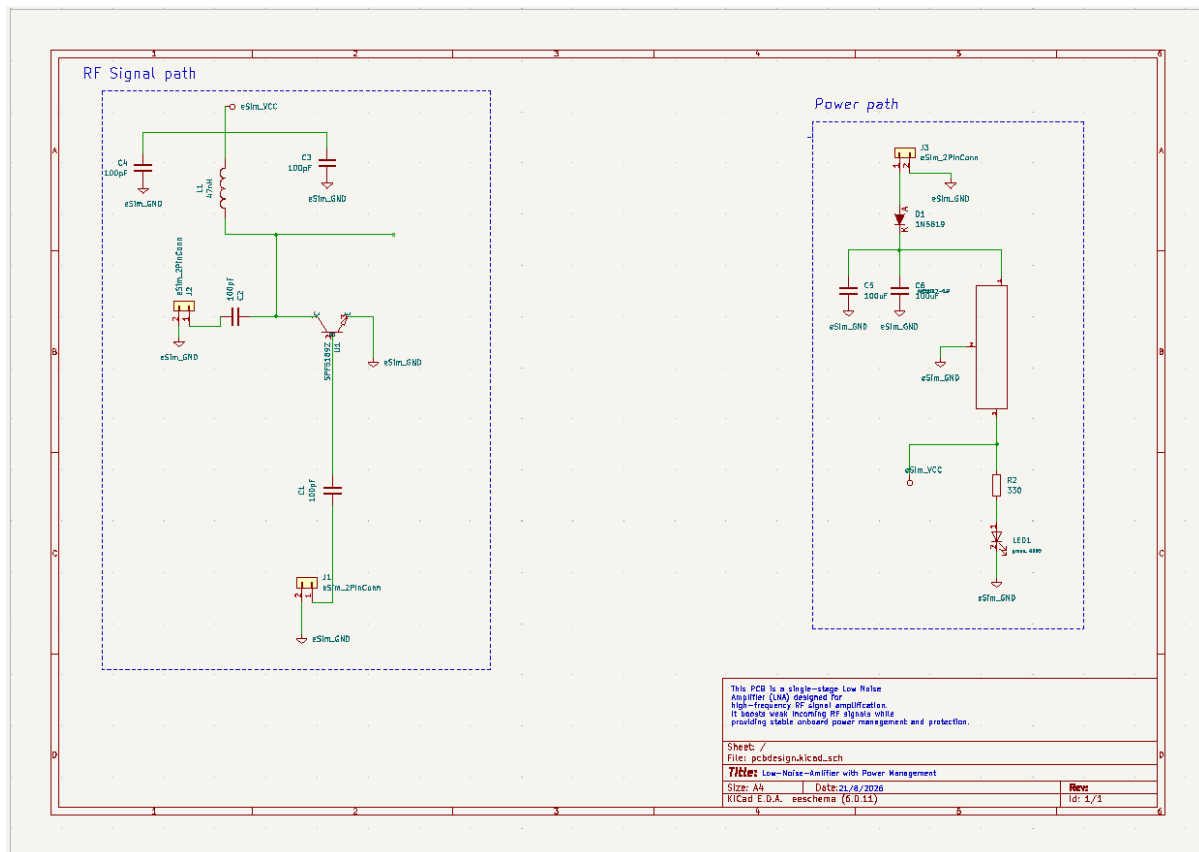
Two edge-launch SMA connectors (J1, J2) provide 50Ω RF input and output access for bench testing with a signal generator and spectrum analyzer. The DC barrel jack (J3) is the sole external power connection.

Status Indicator

A green 0805 LED (LED1) with a 330Ω current-limiting resistor (R2) is tied to the regulated 5V rail, giving a simple visual confirmation that the board is powered.

3. Schematic Screenshots

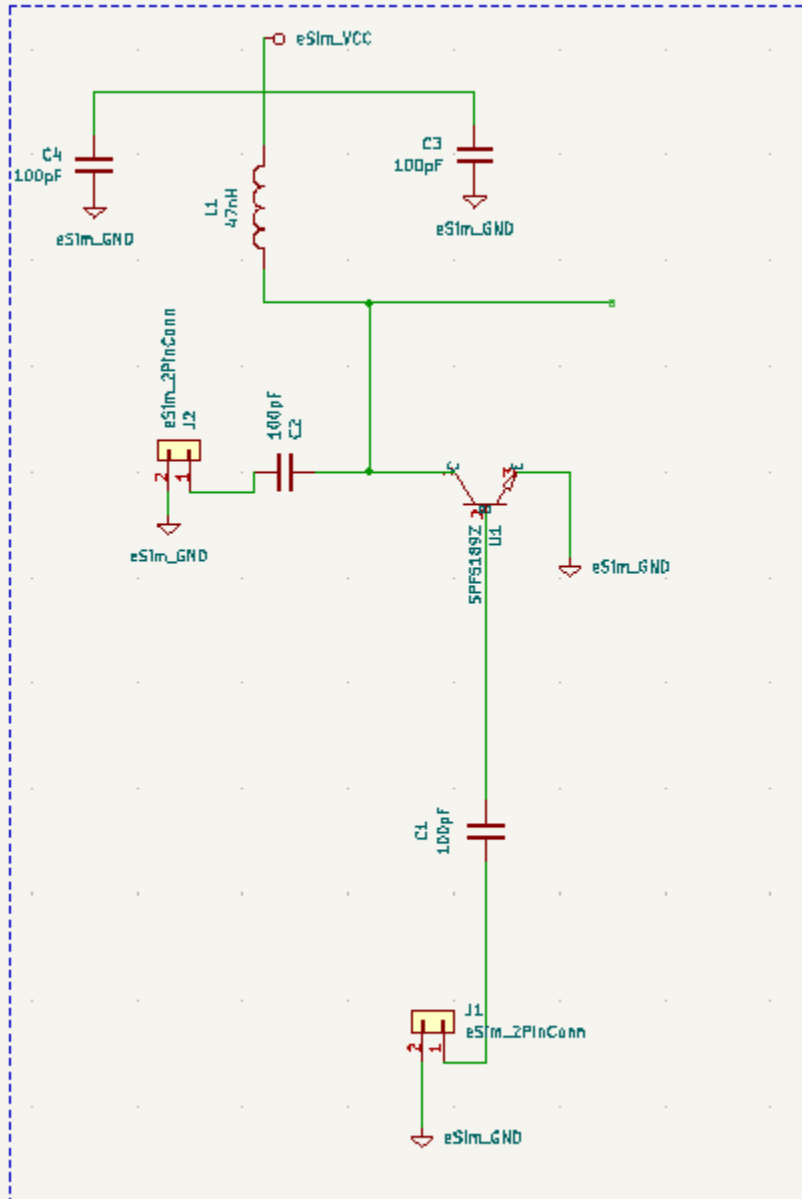
The complete schematic was captured in eSim's eSchema editor



RF Signal Path

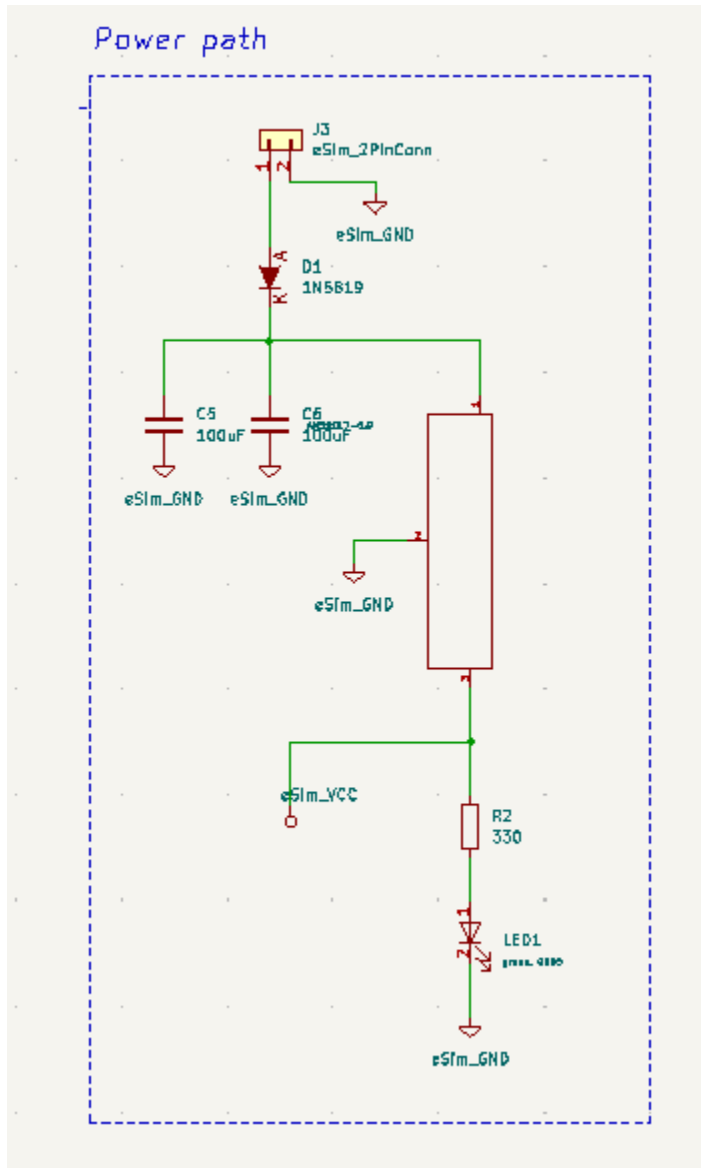
Shows J1, C1, U1, C2, J2 along the RF signal path, and C3, C4, L1 forming the local decoupling and bias tee network at the LNA's combined RF-output/bias pin.

RF Signal path



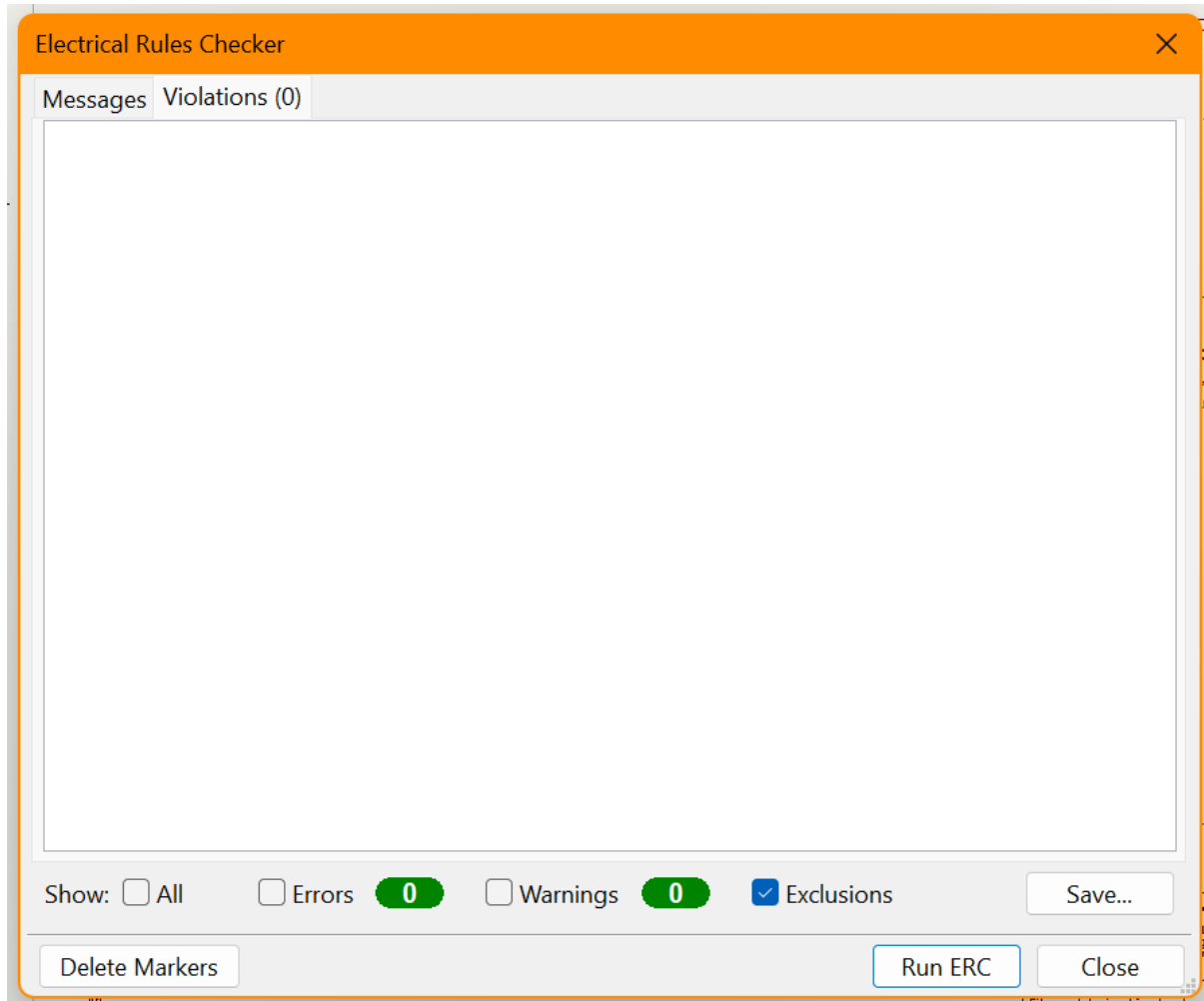
Power and Indicator Path

Shows J3, D1, U2, C5, C6 forming the regulated power path, and R2, LED1 forming the status indicator.



Note: Power and ground nets were connected using esim labels ,so that blocks placed in different areas of the sheet stay readable while remaining electrically connected. ERC was run to confirm every labeled net resolves correctly.

4. Electrical Rules Check (ERC) Result



4. Footprint Assignment

All footprints below are available directly in eSim / KiCad's default libraries.

RF Active Component

- U1 (SPF5189Z): Package_TO_SOT_SMD:SOT-89-3

Passive Components

- Resistors (R2): Resistor_SMD:R_0603_1608Metric
- 0603 capacitors (C1, C2, C3): Capacitor_SMD:C_0603_1608Metric
- 0805 capacitors (C4, C5, C6): Capacitor_SMD:C_0805_2012Metric
- RF choke inductor (L1): Inductor_SMD:L_0603_1608Metric

Connectors

- SMA edge-launch connectors (J1, J2): Connector_Coaxial:SMA_Amphenol_132134_EdgeMount
- DC barrel jack (J3): Connector_BarrelJack:BarrelJack_Horizontal

Power Components

- Reverse-protection diode (D1): Diode_SMD:D_SOD-123
- LDO regulator (U2): Package_TO_SOT_SMD:SOT-223-3_TabPin2

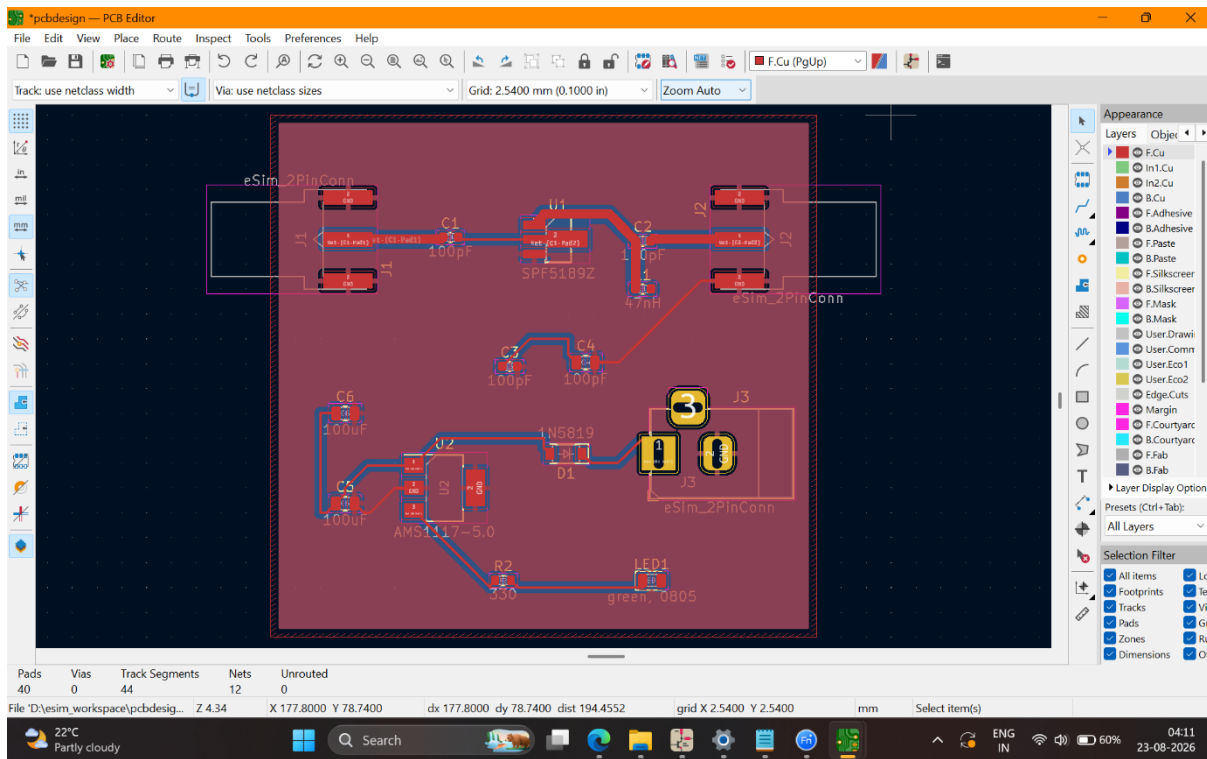
Indicator

- LED1: LED_SMD:LED_0805_2012Metric

5. PCB Layout

The PCB layout was completed in the KiCad PCB editor, accessed through eSim's integration, using a 4-layer stackup.

Complete Board — All Layers View



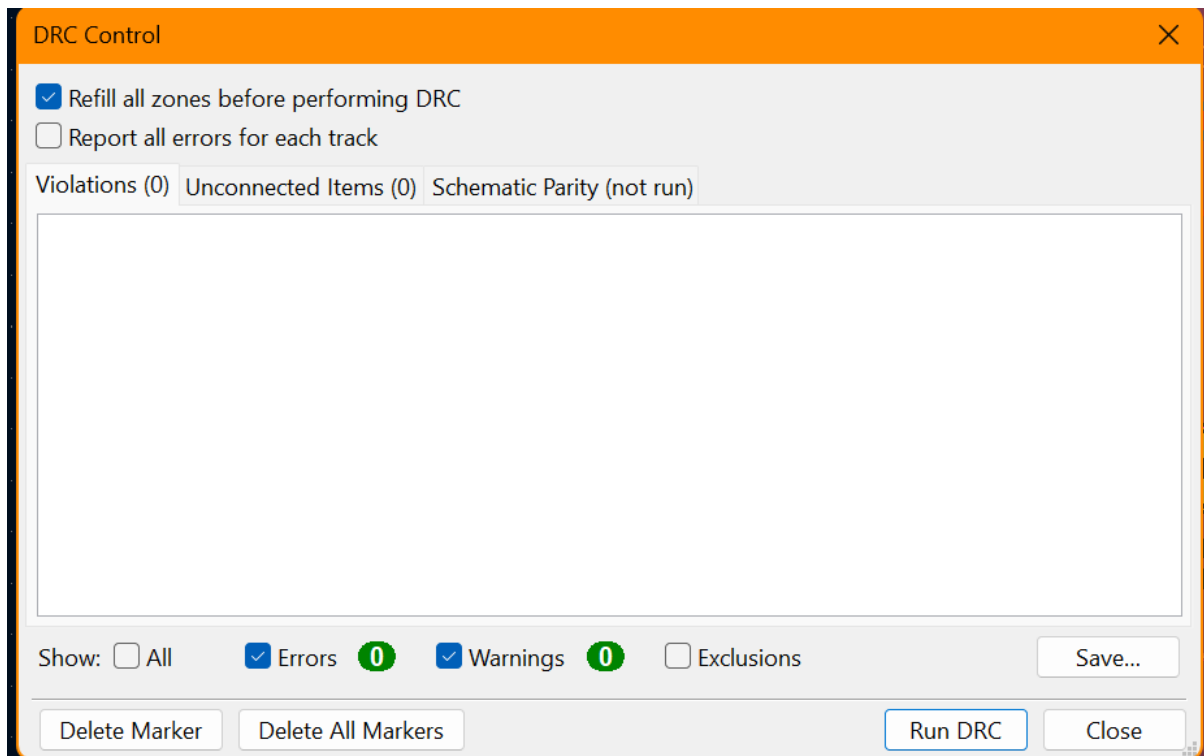
Layer	Name	Type	Purpose
			ground-stitching vias to shield the entire board

Signal Routing Guidelines

- RF trace kept as short and straight as possible between J1 and J2, calculated to 50Ω against the In1.Cu reference plane, with no 90° bends.
- Local decoupling capacitors (C3, C4) placed within a few millimetres of U1's bias pin.
- A double row of ground stitching vias flanks the RF trace (F.Cu, In1.Cu, and B.Cu) for shielding.

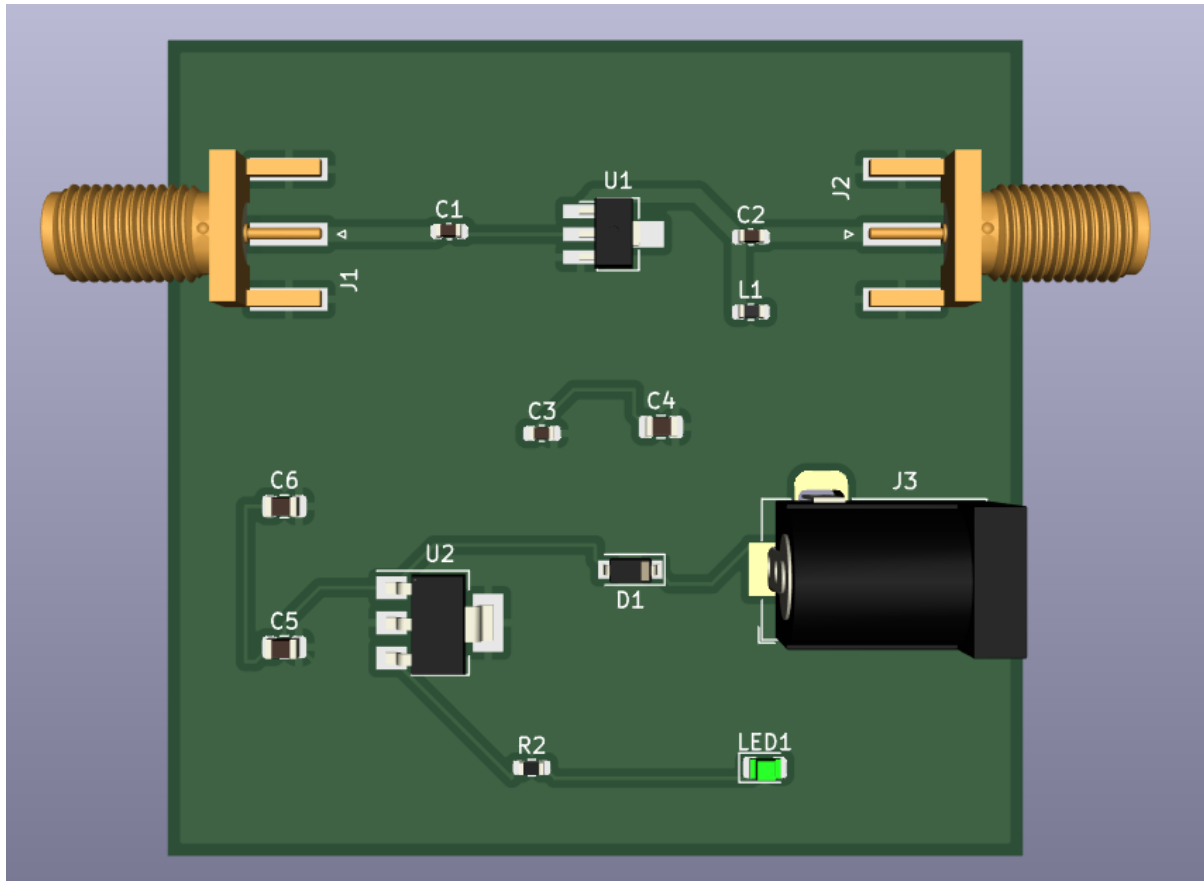
7. Design Rule Check (DRC) Results

DRC was run on the completed layout, with all zones refilled beforehand, to confirm physical manufacturability.



8. 3D Model

The 3D view was generated in KiCad's built-in 3D viewer to verify component placement, footprint accuracy, and overall board appearance before finalizing the design.



9. Conclusion

Power integrity: the dedicated power and ground planes (layers 2 and 3) provide low-impedance distribution, and decoupling at every relevant pin minimizes supply noise reaching the RF stage.

RF signal integrity: keeping the ground plane immediately beneath the RF trace gives a short, predictable return path and a genuinely calculable 50Ω microstrip, rather than an approximation over the full board thickness.

Component sourcing: every part maps to a footprint already present in eSim/KiCad's default libraries, reducing the risk of footprint mismatches during assembly.

1. Fully captured schematic with a clean, verified netlist (target: zero ERC errors).
2. 4-layer PCB stackup with dedicated RF ground and power planes, purpose-built for the RF front-end application.
3. Complete RF signal chain from SMA input to SMA output through a datasheet-proven LNA, with a regulated, isolated bias supply.
4. Manufacturable output: target zero DRC errors, ready for Gerber export.

5. Full documentation: schematic, footprint mapping, layer-by-layer layout, routing, DRC verification, and 3D render, all included in this report.

This design demonstrates schematic capture, RF-aware multi-layer PCB layout, design-rule compliance, and 3D verification — the core competencies the FOSSEE Autumn Internship 2026 screening task is evaluating.

10. Instructions for Execution

Steps to open, verify, and test this project in eSim (Version 2.3 or 2.5):

Opening the Project

1. Install eSim 2.3 or 2.5 from the official eSim download page, if not already installed.
2. Launch eSim and select File → Open Project, then browse to the project folder submitted with this report.
3. Open the schematic file in eSchema to load the full circuit.

Verifying the Design

1. In eSchema, run the Electrical Rules Check (Inspect → Electrical Rules Checker) and confirm 0 errors, 0 warnings.
2. Open the PCB layout in the KiCad PCB editor (via eSim's integration) and run Design Rules Check (Inspect → Design Rules Checker), refilling all zones first, and confirm 0 errors, 0 warnings.

Testing / Reviewing the Board

1. Open the 3D Viewer (View → 3D Viewer) to confirm component placement, footprint orientation, and that no parts overlap or hang off the board edge.
2. Use the layer visibility toggles in the PCB editor to inspect each copper layer individually (F.Cu, In1.Cu, In2.Cu, B.Cu) and confirm the ground and power pours match the layer stackup described in Section 6.
3. Hover over or click individual nets
4. Export Gerber file for manufacturing.